

Earth and Space Science



RESEARCH ARTICLE

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Key Points:

- Unburied distributed acoustic sensing can be feasible on the Moon with suitable cable choice and careful deployment
- Thick unburied cables exhibit flat amplitude and phase spectra, performing comparably to buried cables
- Wind and surface undulations degrade signal quality in unburied terrestrial deployments, especially for thin cables

Supporting Information:

Supporting Information may be found in the online version of this article.

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Controlled Source DAS Coupling Tests: Implications for Unburied Deployment on the Moon and Earth

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Abstract Moonquake recordings are key to understanding the Moon's internal structure, yet existing data from the Apollo missions are limited. Distributed Acoustic Sensing (DAS) offers a promising alternative to conventional seismometers for future lunar seismic monitoring, particularly due to its robustness and suitability for high-scattering environments like the lunar regolith. Since burial of DAS cables may be impractical on the Moon, understanding the coupling performance of unburied cables is critical. In this study, we present experimental results from controlled shaker tests in a lunar regolith simulant, comparing the coupling performance of buried and unburied fiber-optic cables with varying mechanical properties and assessing the impact of wind noise and surface undulations. Our findings show that burial generally improves signal amplitude and phase reliability, while unburied cables yield weaker signals and, in some cases, exhibit phase shifts. However, thicker and stiffer unburied cables achieve comparable performance as buried cables, maintaining stable amplitude ratios and phases. Furthermore, we demonstrate that free-hanging cable segments degrade coupling quality substantially for all tested cables, emphasizing the importance of continuous ground contact. We also show that wind appears to be a major contributor to poor signal quality in unburied DAS deployments on Earth—an issue that would not arise on the Moon due to its lack of atmosphere. These results suggest that with appropriate cable selection and deployment strategies, unburied DAS is a viable option for lunar seismic applications.

Plain Language Summary Understanding the Moon's interior is important for planetary science, and seismic measurements are one of the most effective tools for this purpose. These measurements rely on detecting moonquakes: vibrations that travel through the lunar subsurface. However, the only available moonquake data were collected during the Apollo missions using a small number of traditional seismometers. Distributed Acoustic Sensing (DAS) is a newer technique that could improve future lunar seismic monitoring. DAS uses a fiber-optic cable to measure ground motion along its entire length, effectively turning the cable into thousands of vibration sensors. This method offers dense spatial coverage and could simplify deployment compared to conventional instruments. On Earth, DAS cables are typically buried to ensure good contact with the ground. However, on the Moon, burial may not be feasible. To investigate whether unburied cables can still provide reliable data, we conducted controlled experiments using a lunar soil simulant and several types of fiber-optic cables. We found that thicker, stiffer cables can record stable signals even when unburied, provided they maintain contact with the ground. These results suggest that DAS could be a viable method for lunar seismic sensing, enabling large-scale measurements without the need for complex installation procedures.

1. Introduction

Currently, the only seismic data set recorded on the Moon originates from the Apollo missions, during which both active and passive seismic experiments were conducted. Five stations deployed on the near side of the Moon recorded over 12,000 seismic events from 1969 to 1977 (see Nunn et al., 2020 for a review of the instrumentation and collected data). This data set has led to the identification of four distinct types of natural seismic sources on the Moon, namely, deep moonquakes, shallow moonquakes, thermal moonquakes, and meteoroid impacts (Nakamura et al., 1982). The collected seismic data have been the primary source for estimating the lunar seismic velocity structure and have been extensively analyzed over the decades and continue to receive fresh scrutiny, for example, by Sollberger et al. (2016), who used spatial wavefield gradients to construct an elastic model of the shallow lunar subsurface. A major challenge in interpreting moonquake seismograms is the presence of long-duration and high-amplitude coda waves that are much more pronounced than exhibited in earthquake

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seismograms. Caused by the strongly scattering and low attenuating lunar shallow subsurface, the coda waves obscure later arriving seismic phases.

An approach that could help address this challenge is an emerging sensing technology called Distributed Acoustic Sensing (DAS), which measures dynamic strain over long distances (e.g., Hartog, 2017; Lindsey & Martin, 2021). DAS has rapidly gained traction in a broad range of geophysical applications as an alternative to conventional ground motion sensors such as geophones and seismometers. A DAS setup consists of a fiber-optic cable connected to an interrogator. The interrogator sends laser pulses into the optical fiber and measures the returning light that experiences Rayleigh back-scattering from naturally occurring inhomogeneities in the fiber smaller than the probing wavelength. This backscattered light undergoes phase changes as the scatterers move due to strain on the fiber, consequently, the dynamic strain of the optical fiber can be inferred, thereby transforming the optical fiber cable into a dense array of vibration sensors. As a result, DAS is sensitive to a length change of the cable which—in the case of perfect coupling of cable to the ground—corresponds to the inline strain.

DAS is a technology that is robust and easy to deploy, making it a promising candidate for lunar deployment. A single fiber-optic cable attached to an interrogator can record over tens of kilometers and replace thousands of sensors, eliminating the need for individual power supplies or data acquisition systems. Fiber-optic sensors can withstand harsh environmental conditions, including extreme temperatures and radiation. They have been successfully deployed in environments ranging from cryogenic temperatures (Yang et al., 2021) to temperatures as high as 1,000°C in ϕ -OTDR-based DAS experiments (Xu et al., 2025). Wu et al. (2021) demonstrated that OFDR-based fiber sensors maintained stable Rayleigh backscattering profiles over multiple months inside a nuclear reactor core, exposed to both high temperatures and neutron radiation. Additionally, long-duration space exposure experiments have shown that fiber-optic cables can remain functional after years in orbit, even when unshielded (e.g., Hartmayer et al., 1993). This study confirms that cables can endure temperature cycling, radiation exposure, and polymer aging in space. However, planned mission duration is critical since radiation effects and material degradation accumulate over time, and the small but non-negligible risk of micrometeoroid impacts on the cable and a resulting fiber breakage increases with exposure time. Recent modeling studies have demonstrated the potential of DAS for lunar applications (Wu et al., 2024). The dense channel spacing that DAS offers is particularly suited for high-scattering environments such as the lunar shallow subsurface and could allow DAS to recover deep seismic phases that were obscured in Apollo data due to scattering. Zhai et al. (2024) estimate that DAS could detect around 60% of the moonquakes observed by the Apollo seismometers. However, with foreseeable improvements in DAS technology this number could rise to detect nearly all moonquakes previously observed by Apollo. Their findings emphasize the critical role of DAS performance in moonquake detectability, underscoring the relevance of understanding strain transfer and amplitude behavior.

However, several open questions need to be addressed before DAS can be deployed on the Moon. A key challenge is determining how to best deploy the fiber-optic cable ensuring good coupling. On Earth, burial is commonly used to ensure good coupling, and studies have shown that it significantly improves signal quality compared to unburied deployment (Dou et al., 2017; Rossi et al., 2022). However, burial may be infeasible in the lunar environment. Unburied deployments on Earth often exhibit poor signal quality, which has been partly attributed to atmospheric acoustic noise, particularly wind and thunder (Carr et al., 2025; Hudson et al., 2021; Viens et al., 2025). This atmospheric noise is not expected to be a significant source of noise, as the Moon has only a thin exosphere. Other potential noise sources exist on the Moon, such as regolith transport or micrometeorite impacts, but these are intermittent phenomena. Regolith transport, for instance, occurs primarily around lunar sunrise and sunset due to electrostatic charging effects (Colwell et al., 2007), and is therefore unlikely to constitute the same continuous noise as even a light wind does in terrestrial environments. However, it is still unclear whether the poor performance of unburied fiber is primarily due to poor ground coupling or rather due to atmospheric noise. Understanding this distinction is critical for assessing whether unburied deployment on the Moon can yield satisfactory seismic data. Recent studies have begun to address coupling in buried and unburied DAS deployments (e.g., Spikes et al., 2019). Harmon et al. (2022) investigated surface coupling strategies in a grassy field, showing that even uncoupled fiber can record near-field seismic energy, though pinned and weighted configurations significantly improved signal-to-noise ratios. They also derived an empirical response function and emphasized the need for systematic tests across substrates and coupling methods. Harmon et al. (2024) extended this work by demonstrating DAS feasibility at lunar analogue sites in volcanic terrain, using weighting and shallow-burial strategies and comparing DAS to geophones and broadband seismometers. Hudson et al. (2024) introduced a coherence-based metric to identify poorly coupled segments. While these studies

provide valuable insights, controlled experiments isolating coupling effects, particularly in the absence of atmospheric noise, are still needed to guide optimal strategies for lunar DAS deployments.

This study employs lunar regolith simulant (LRS) to examine the relationship between the conditions of a fiber-optic cable deployment and data quality, highlighting implications for lunar exploration. We are building upon the work of Zandanel et al. (2026) which used the same experimental setup to record local and regional earthquakes and found that unburied fiber is capable of detecting earthquakes with a weaker amplitude than buried fiber, but with only a modest decrease in the signal-to-noise ratio. This study aims to quantify this effect and investigate whether the observed amplitude attenuation is frequency-dependent and if the phase recorded with unburied fiber is reliable. Using a shaker source to produce controlled and repeatable ground motion in the LRS, we examined the coupling of different cable types with varying mechanical properties, such as outer diameter and weight. Employing cables with drastically different properties is motivated by two key considerations: (a) comparing the performance of cables with differing characteristics can help elucidate the physical mechanisms that govern fiber-regolith coupling and how these mechanisms depend on cable properties, and (b) for lunar deployment, weight and robustness are critical factors. Due to the high cost of transport to the Moon, lightweight and robust cables are highly desirable, especially given that deployment strategies are still under development, with potential concepts of launching the cable onto the surface. Additionally, we investigated the impact of ambient wind to understand how much wind noise contributes to the poor signal quality of unburied fiber observed on Earth. Lastly, we investigated the impact of small-scale surface topography on coupling quality to assess the robustness of unburied cables to surface heterogeneities.

2. Methods

2.1. Experimental Setup

An experiment was built that mimics some of the conditions that are present on the Moon. A 9.5-m-long trough was filled with a generic lunar regolith simulant (LRS), composed of crushed basalt (Figure 1c). The trough was filled to a depth of approximately 59 cm with the LRS, forming a flat surface. The LRS consists of 35%–50% plagioclase, 10%–45% pyroxene, 10%–15% olivine, 5%–10% calcite, 3%–7% magnetite, and up to 7% hematite (in mass%) (Zandanel et al., 2026). The grain size distribution, determined by sieving, consists of 35% (by mass) of particles larger than 2.38 mm (with very few exceeding 10 mm), 26% between 1 and 2.38 mm, 11% between 0.5 and 1 mm, and 28% smaller than 0.5 mm.

Systematically investigating the coupling behavior of various cable types and deployment configurations could only be achieved by exciting the LRS with controlled frequencies and amplitudes. To generate such controlled ground motion, a shaker (Brüel & Kjær 4809) was mounted at the short end of the trough, aligned with the LRS surface, and oriented to excite motion along the longitudinal axis (see Figure 1d). The shaker was driven with a sequence of single-frequency signals ranging from 3 to 30 Hz in 1 Hz increments. Each frequency was excited for 250 periods and repeated at 10 different amplitude levels. Frequencies below 3 Hz were excluded due to shaker limitations, and frequencies above 30 Hz were excluded to avoid the resonance of the trough, which had the first mode at approximately 30 Hz (see Figure 1b). The energy of most natural moonquakes is below 20 Hz (Civilini et al., 2023). For illustrative purposes, an earthquake was simulated by driving the shaker with a real earthquake recording as input. The driving signal was taken from a magnitude (M_w) 4.3 earthquake near Trinidad, Colorado on 10 March 2023 recorded on the seismometer buried in the lunar simulant labeled Event 2 in Zandanel et al. (2026).

We collected DAS data with three cables that exhibit strongly differing mechanical properties: a tight buffered cable with an outer diameter (OD) of 5.5 mm (referred to as OD5.5), a tight buffered cable with an OD of 3 mm (OD3), and a jacketed cable with an OD of 0.9 mm (OD0.9). The properties of those cables are summarized in Table 1.

To investigate the impact of burial on coupling, we deployed the cables at three depths in the LRS: unburied (laid directly on the surface without additional coupling), shallow-buried (at a depth of approximately 2–3 cm below the LRS surface), deep-buried (at a depth of approximately 11 cm) (see Figure 1a). Each cable type was deployed in both unburied and shallow-buried configurations. For the OD0.9, two separate cables were deployed in parallel at each depth to provide repeatability. The OD5.5, which contains two optical fibers within a single cable, was interrogated on both fibers, yielding two measurements from the same physical cable. The signals from both

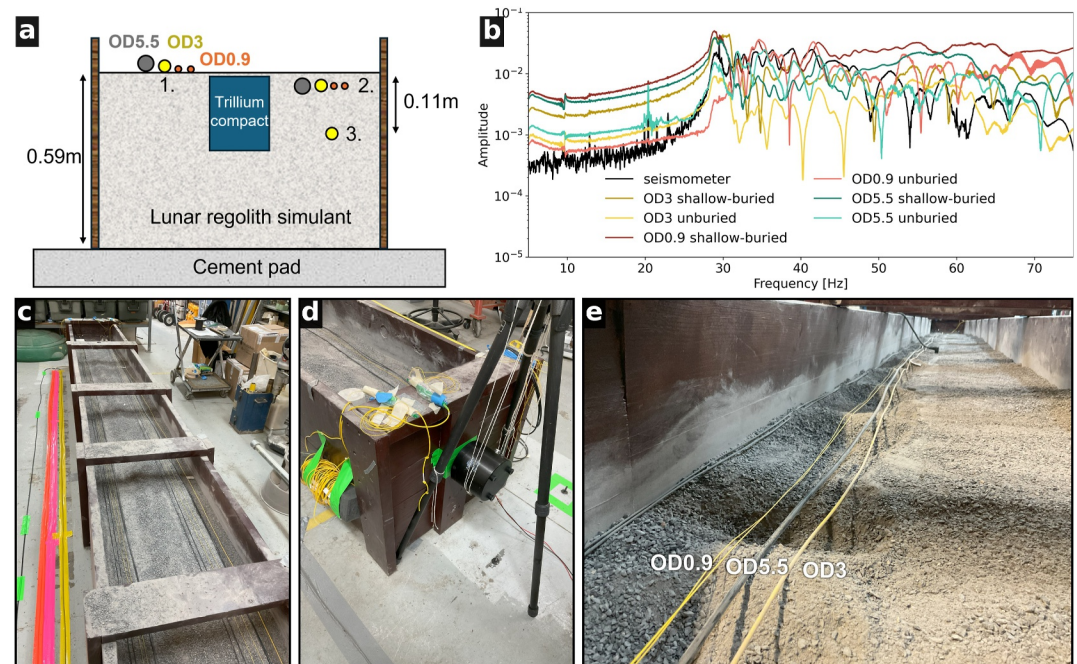


Figure 1. Overview of the experimental setup and conditions. (a) Schematic cross-section of the trough showing the three deployment depths of the fiber-optic cables (1) unburied, (2) shallow-buried, and (3) deep-buried. (b) Amplitude spectrum of a shaker sweep recorded on different fiber segments with minimal undulations, with excitation ranging from 5 to 150 Hz. Resonance behavior present at frequencies above 30 Hz. For comparison, the amplitude spectrum of the co-located seismometer is also shown (normalized for visual clarity), illustrating that the resonance is a property of the experimental trough rather than the fiber coupling. (c) Photograph of the 10-m-long trough filled with lunar regolith simulant (LRS) and deployed cables. (d) Shaker mounted at the short end of the trough for controlled excitation. (e) Surface undulations created by grain piles (spaced 15–25 cm apart), with cables suspended between contact points to simulate irregular topography.

fibers within the same cable were very similar, which is expected given that they share the same cable and thus experience the same ground-cable coupling. Consequently, only one fiber per cable is presented in the results section for clarity. Only the OD3 was additionally deployed in the deep-buried configuration. After deployment, all cable segments were spliced together into a single continuous chain. This allowed the entire set of segments to be connected to one interrogator and recorded simultaneously during the experiment. To test the robustness of the results and assess the sensitivity of coupling to small variations in deployment, all unburied cables were redeployed after one round of data acquisition. This involved removing and repositioning the fibers on the surface, followed by a repeated acquisition under the same conditions.

DAS data were acquired using a Terra15 Treble + interrogator at a sampling rate of 3 kHz, with a gauge length and pulse length of 2.45 m and a channel spacing of 0.82 m. This configuration was chosen to balance spatial resolution and signal-to-noise ratio. For comparative analysis of the cables and deployment types, we used the central DAS channels at the midpoint of the trough.

Table 1
Specifications of the Used Fiber-Optic Cables

Cable type	Abbrev.	OD (mm)	Nr. of cores	Buffering	Weight
PS-2S2M-1PU055-01	OD5.5	5.5	2 SM, 2 MM	Tight buffered	27 g/m
SMF-28-100	OD0.9	0.9	1 SM	Yellow Hytrel tubing	3.1 g/m
Silixa	OD3	3	1 SM	Tight buffered	8.6 g/m

Note. OD, outer diameter; SM, single-mode fiber; MM, multi-mode fiber. Cable OD5.5 contains both single-mode and multi-mode fibers, but only the single-mode fibers were interrogated for the measurements.

A seismometer (Trillium compact 120s) served as a ground-truth reference and was placed at the center of the trough, aligned with the DAS channels and sampled at 250 Hz.

While the trough geometry influences the wavefield, including potential waveguide effects and dispersion, all cables and the seismometer were deployed close together within the same trough and thus experience the same wavefield, ensuring that our comparative analysis remains valid. The trough was located indoors, shielding it from natural wind. To simulate wind noise, a fan was positioned approximately 30 cm above the cables at the center of the trough, blowing vertically downward. The fan operated at three intensity settings, corresponding to maximum wind speeds of 2, 2.4, and 3 m/s at a distance of 30 cm away from the fan. Those wind speeds, classified as a “light breeze” on the Beaufort wind scale, are typical of outdoor conditions on Earth.

To assess the impact of surface irregularities, we introduced artificial undulations in the trough by forming grain piles of varying spacings (15–25 cm, 30–50 cm, 60–100 cm) with a height of approximately 5 cm, sufficient to keep the cables suspended between contact points. Cables were laid across these piles such that they contacted the ground only at the peaks and were suspended in between (see Figure 1e). The shaker experiment was repeated for each undulation configuration.

2.2. Data Processing

To compare the fiber segments, we analyzed the DAS signal recorded at the central position of each segment. A tap test was performed at the midpoint of the trough to identify the central positions, which were localized with an estimated accuracy of half the channel distance. Since signal amplitudes vary strongly along the trough, precise positional comparison is critical. Channel positions do not necessarily coincide between segments, so comparing the nearest channel to the midpoint for each segment would introduce positional errors. Therefore, we interpolated the signal at the exact midpoint by computing a weighted average of the two adjacent channels for each segment (see Supporting Information S1 for geometric illustration). The amplitude and phase behavior of each fiber segment were characterized as a function of both excitation frequency and amplitude of the shaker. For each excitation frequency, the corresponding time window was divided into 1-s segments. The frequency spectra of these segments were computed and averaged to obtain a single spectrum, ensuring consistent spectral resolution and comparability across all excitation frequencies. From each averaged spectrum, the amplitude and phase at the excited frequency were extracted. The seismometer data were corrected by removing the instrument response and then processed using the same spectral analysis workflow as the DAS data.

This procedure allowed us to investigate how the recorded amplitude, and thus the coupling quality between the fiber and the ground, varies with cable type, deployment depth, strain amplitude, and frequency. The amplitudes were normalized by the amplitudes recorded by the collocated seismometer to account for the frequency-dependent response of the shaker, yielding amplitude ratios. The seismometer data served as the ground-truth reference because they were corrected for the known instrument response, resulting in a flat amplitude and phase response across the tested frequency range. Literature suggests that DAS also exhibits flat amplitude and phase spectra when well coupled (Paiz et al., 2021), and this assumption can be verified through comparison with the seismometer. These amplitude ratios are analogous to an instrument response function, however, they represent a comparative metric rather than a true response, because DAS measures strain rate while the seismometer measures ground velocity. Consequently, amplitude ratios are not expected to be unity even under ideal coupling conditions. Theoretically, conversion approaches exist to convert DAS recordings to velocity (e.g., Trabattori et al., 2023). However, such conversions assume ideal coupling and long linear cable segments - conditions not met in our experiment. Therefore, we refrain from converting DAS and the seismometer to the same units and instead perform a qualitative comparison using amplitude ratios. The absolute maximum ground particle velocities recorded by the seismometer were on the order of 10^{-4} m/s. Illustration of the maximum velocities across frequencies and shaker amplitude levels is included in the Supporting Information S1.

To summarize the coupling quality of each cable and deployment configuration with a single metric, we computed the average amplitude ratio. This metric was obtained by averaging the amplitude ratios across all excitation frequencies and amplitudes. Additionally, for the OD5.5 and OD0.9 where two fibers of each cable could be interrogated at the same point—either from two fibers within the same cable (OD5.5) or from two separate cables (OD0.9)—the results from both measurements were averaged to obtain one representative value. An exception is OD0.9 unburied for which the results from both fibers were processed separately, since their performance was very variable, as discussed in more detail later on. The uncertainty of the average amplitude ratio

was estimated by accounting for the localization uncertainty of the central channel. Specifically, the analysis was repeated after shifting the central position by ± 0.5 channel spacing, and the resulting variation in amplitude was used to quantify the uncertainty. This channel location uncertainty was considered the dominant source of error, as the shaker was mounted at one end of the trough, resulting in a strong amplitude gradient along the trough axis. Consequently, even small deviations in channel position could lead to significant differences in recorded amplitude.

For the analysis of wind-induced noise, we also focused on the DAS signals recorded at the midpoint of the trough where the fan was positioned. The wind noise data were processed by calculating the power spectral density (PSD) of each trace using Welch's method, with a window length of approximately 3 s. This allows us to assess the wind's impact across different frequency bands. To quantify the average intensity of wind-induced vibrations in the seismic frequency band, each trace was further filtered with a fourth-order acausal Butterworth low-pass filter with a cutoff frequency of 20 Hz. The mean absolute amplitude of the filtered signal was then computed in the time domain, providing a representative measure of the sub-20 Hz strain rate noise for each cable segment and wind condition. Here again, unburied cables were redeployed after one round of data acquisition.

3. Results

3.1. Comparison of Cable Types and Deployment

Figures 2a and 2b illustrate how each cable and deployment type responded to shaker excitation, in terms of phase and amplitude. Shallow and deep-buried fiber segments exhibit relatively constant, frequency-independent amplitude ratios. Their phases are flat and consistent with the seismometer, indicating reliable strain measurements in the tested frequency-range aside from a uniform amplitude scaling. Unburied fiber segments OD5.5 and OD3 also exhibit stable amplitude ratios and flat phases with the exception of an observed phase shift at 25 Hz for OD5.5. Unburied OD0.9 exhibits considerable variability in coupling quality across multiple observations, including the two deployed segments and repeated deployments. Some observations showed flat amplitude and phase behavior, while others exhibited erratic amplitude and phase variations. In Figure 2, both OD0.9 unburied fiber segments are shown separately to highlight the variability observed between deployments. Repeatability and deployment sensitivity are further discussed in Section 4.1. For one of the unburied OD0.9 segments, a pronounced phase shift is observed at frequencies above 13 Hz, which may indicate that the fiber is resonating at those frequencies.

A comparison across different shaker amplitude levels shows a linear relationship between input and recorded amplitude, with no evidence of nonlinear behavior (see Supporting Information S1). Figure 2c shows the average amplitude ratios across all frequencies and amplitudes. For buried cables, the values are relatively similar, with OD0.9 showing the highest amplitude ratios, but still close to the uncertainty range of the other cable types. In contrast, unburied cables generally exhibit lower amplitude (roughly half that of buried cables), except for OD5.5, which shows comparable values in both configurations.

In addition to amplitude and phase analysis, we computed the signal-to-noise ratio (SNR) for each fiber segment, shown in Figure 2d. The SNR was calculated by comparing the signal amplitude during excitation to the noise amplitudes. The SNR trends largely mirror the average amplitude ratios, but with some notable differences. OD5.5 consistently shows higher SNR than OD3 in both buried and unburied configurations. Unburied OD3 achieves SNR values comparable to its buried counterpart, while OD0.9 exhibits significantly lower SNR when unburied.

3.2. Impact of Surface Undulations

In field deployments, ensuring that fiber-optic cables lie flat and maintain continuous ground contact is labor-intensive. Faster, less controlled deployments may result in cables spanning small surface undulations, potentially degrading coupling quality. To investigate this effect, we introduced artificial surface undulations of varying sizes in the trough. In these deployments, the cables are only in contact with the LRS at the peaks of the undulations. Figure 3 illustrates the impact of these undulations on the signal quality. Coupling quality decreases significantly when cables are suspended across undulations rather than in continuous contact with the ground. For small undulations, the OD0.9 already shows a marked reduction in coupling quality, expressed as lower amplitudes and loss of flat amplitude and phase spectra across all tested frequencies. In contrast, the OD5.5 and the

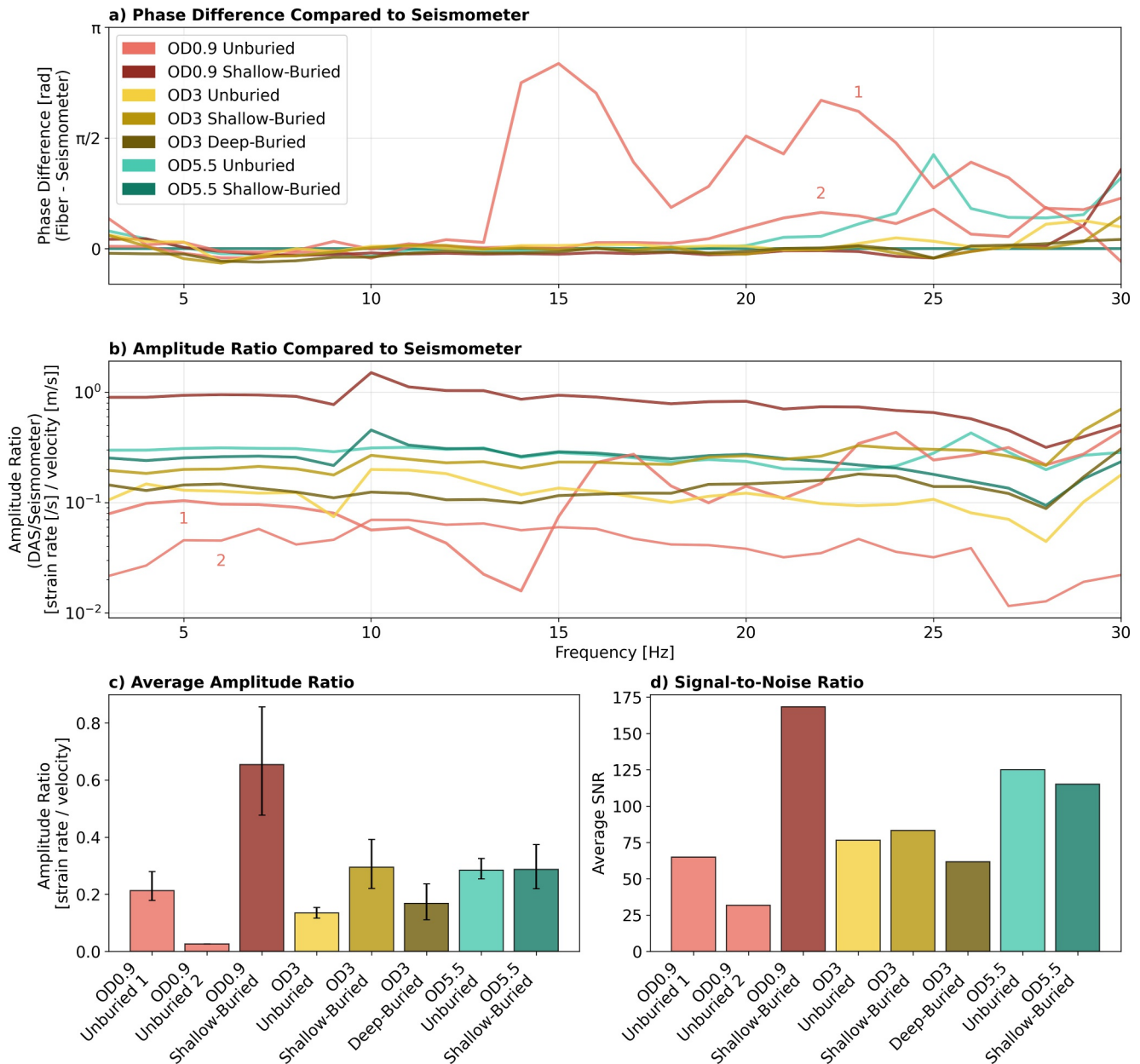


Figure 2. Fiber-optic cable coupling quality for different cable types and deployment configurations. (a) Phase difference of DAS phase relative to seismometer phase (shaker amplitude level 7). (b) Amplitude ratio of DAS amplitude compared to seismometer amplitude across frequencies 3–30 Hz (shaker amplitude level 7). (c) Coupling quality summary showing amplitude ratios averaged across all excited frequencies and amplitudes, with error bars indicating variability due to channel location uncertainty. (d) Signal-to-noise ratio of fiber segments averaged over all excited frequencies and amplitudes. For the OD0.9 unburied configuration, both deployed fibers are shown in all subfigures to illustrate the observed variability between deployments. In contrast, other cables with multiple recordings showed close agreement and are represented by a single curve or value.

OD3 maintain relatively good coupling at frequencies below 10 Hz but also exhibit poor performance at higher frequencies. For larger undulations, all unburied cables show substantial signal degradation, characterized by low amplitudes and strong variability. Additionally, phases become erratic, with large and inconsistent phase shifts. These drastic phase shifts, especially at higher frequencies may indicate fiber resonance. Notably, the onset frequency of these phase shifts decreases as the undulation spacing increases, suggesting that longer free-hanging segments lower the resonance frequency.

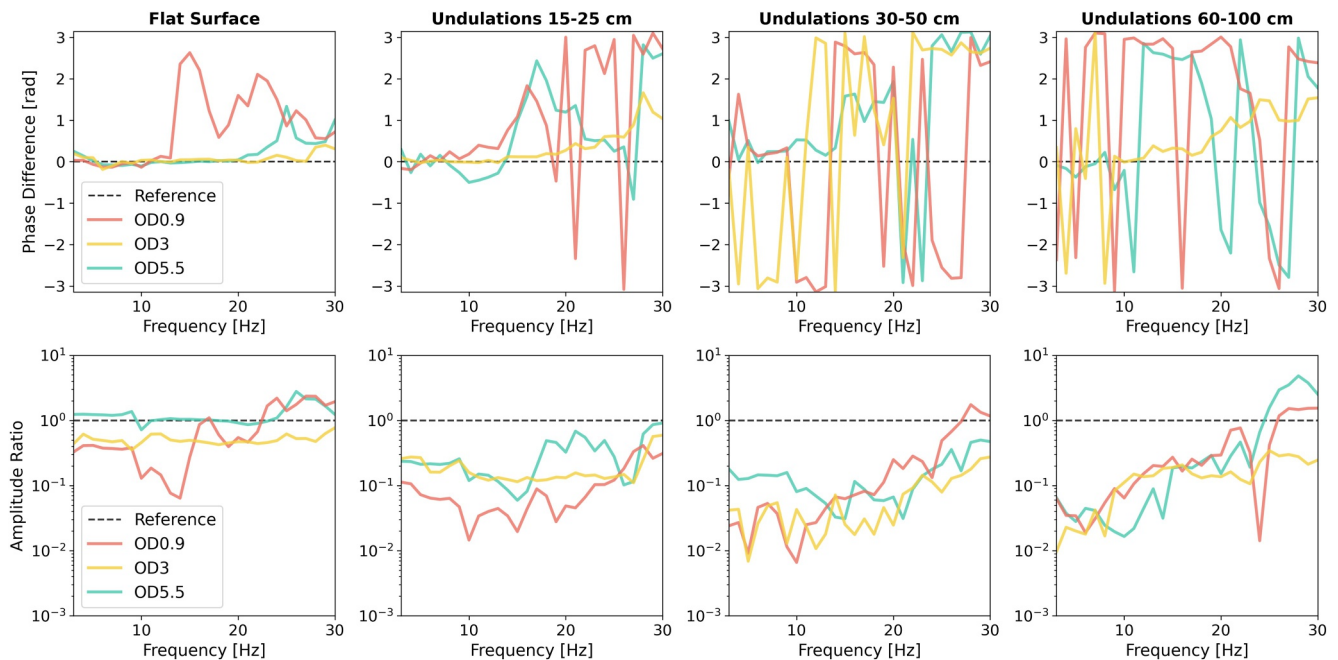


Figure 3. Impact of surface undulations on the coupling of unburied cables (shaker amplitude level 7). Top row: phase difference figures; bottom row: amplitude ratio figures. The leftmost column represents the flat surface case (as in Figures 2a and 2b), while the subsequent columns show increasing undulation spacing. All plots compare unburied cables to the shallow-buried OD5.5 cable (dashed horizontal line), which serves as the reference for amplitude and phase stability. As undulation spacing increases, coupling quality progressively degrades, evidenced by reduced amplitudes and phase shifts, particularly in the two cases with the larger undulations.

3.3. Impact of Wind Noise

To assess the influence of wind on DAS signal quality, we compare the noise floor of fiber segments exposed to artificially generated wind with those not exposed. Since the trough is located indoors, the fibers are naturally protected from ambient wind, allowing for controlled testing. Wind noise might contribute significantly to the poor signal quality observed in unburied cables deployed outdoors. To quantify this effect, we used a fan to simulate wind and analyze the resulting increase in noise levels. The results are presented in Figure 4 (for improved visibility, the spectra are only shown up to 200 Hz). The shallow-buried fibers are practically unaffected by wind, confirming their robustness against wind noise. In contrast, all unburied fibers exhibit elevated noise levels, though the degree of impact varies significantly between cable types. OD5.5 is the least affected, with wind-induced noise primarily below 200 Hz and maximum amplitudes of the wind noise reaching approximately -130 dB/Hz. OD3 shows increased sensitivity, with noise extending up to 400 Hz and peak amplitudes around -125 dB/Hz. OD0.9 is the most affected, with wind noise up to 700 Hz. Particularly strong wind amplitudes are observed in bands between 20–30 Hz and 400–600 Hz, with maximum amplitudes reaching -110 dB/Hz. For seismological applications, the frequency band below 20 Hz is of particular relevance. In this band, the wind-induced noise amplitudes at fan speed 3 are 1.99×10^{-7} m/m/s for OD5.5, 7.15×10^{-7} m/m/s for OD3, and 2.57×10^{-6} m/m/s for OD0.9, sufficient to obscure earthquake signals that would otherwise be well above the detection threshold in the absence of wind noise.

To illustrate the impact of wind noise on seismic signal detectability, we recorded a shaker-generated simulated earthquake with and without wind noise from the fan. As shown in Figure 5, the presence of wind noise significantly affects the signal detectability. On OD0.9 and OD3 unburied, the event becomes undetectable while on OD5.5 unburied, the signal-to-noise ratio is noticeably reduced. The PSDs of the recordings show similar noise patterns as Figure 4 and are included in the Supporting Information S1.

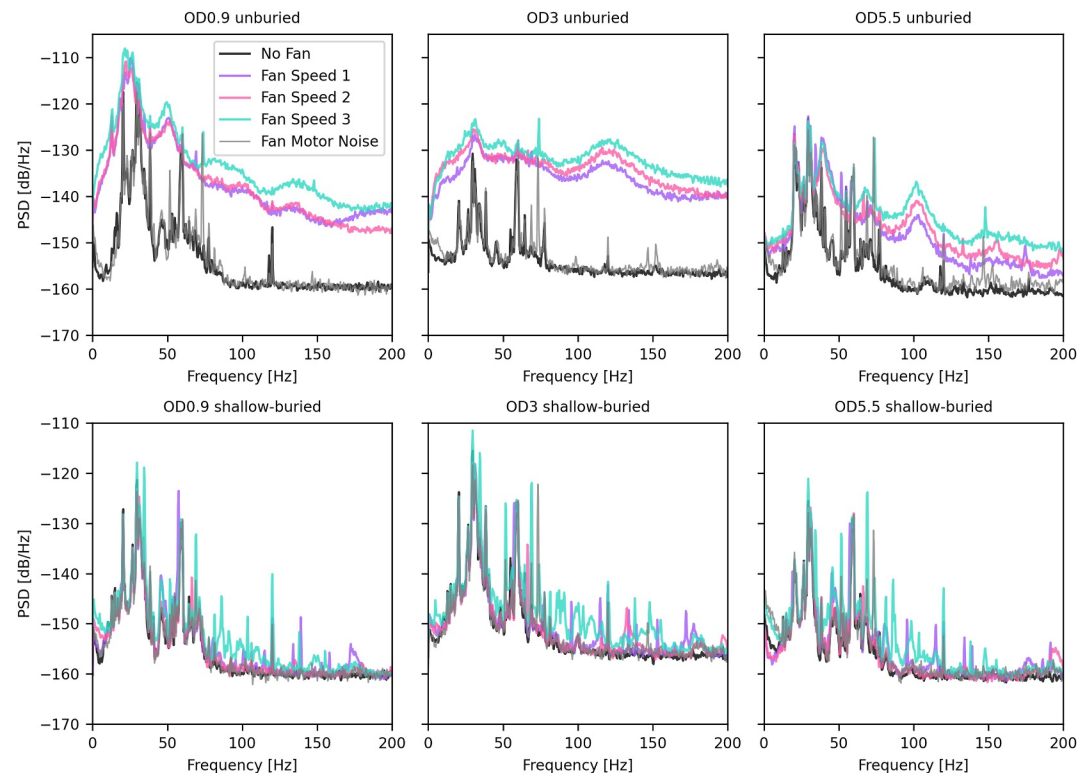


Figure 4. Power spectral density (PSD) comparison of wind-induced noise across different cable segments. Top row: unburied cables (OD0.9, OD3, OD5.5); bottom row: corresponding shallow-buried segments. Each plot shows PSD curves from 0 to 200 Hz for the four conditions. “No Fan” corresponds to data acquisition in the absence of any fan operation, “Fan Speed 1 to 3” correspond to the fan blowing directly onto the cables at wind speeds of 2, 2.4, and 3 m/s, respectively, while the “Fan Motor Noise” condition was recorded with the fan operating but directed away from the cables. This setup captures the ambient noise and fan motor noise without wind impact on the fibers. The difference between the “Fan Motor Noise” and “Fan Speed” curves isolates the effect of wind on signal quality.

4. Discussion

4.1. Discussion of Results

Buried cables consistently recorded signals with correct phase and stable amplitude ratio (Figures 2a and 2b). Frequency-independent amplitude ratios and zero phase differences indicate good coupling—waveforms are recorded faithfully with only a constant amplitude scaling. In contrast, frequency-dependent variations or phase shifts indicate uncorrectable waveform distortion. Larger amplitude ratios also reflect more efficient strain transfer, resulting in improved signal-to-noise ratios. Amplitude ratios between cable types differed within a factor of two. The deep-buried cable showed slightly lower amplitude ratios than the shallow-buried ones, but within uncertainty (Figure 2c). Since only one deep-buried cable was tested, no definitive conclusions about burial depth can be drawn.

Cable type had a more pronounced impact in unburied deployments. The two thicker cables (OD5.5 and OD3) recorded amplitudes comparable to their buried counterparts and maintained reliable phase information across most frequencies. The thinner OD0.9 cable exhibited significantly lower amplitudes and high variability across different deployments. After redeployment for repeated testing, the OD0.9's signal quality degraded noticeably. We observed that the initial surface had more fine grains and dust than after reworking. Additionally, in the initial deployment, the cables had been in place for a few months before data acquisition, allowing them to settle slightly into the fine surface dust, which likely improved coupling. After redeployment, cables were slightly lifted by their stiffness, reducing contact and coupling. These observations suggest that thin cables are particularly sensitive to small variations in deployment geometry and surface conformity.

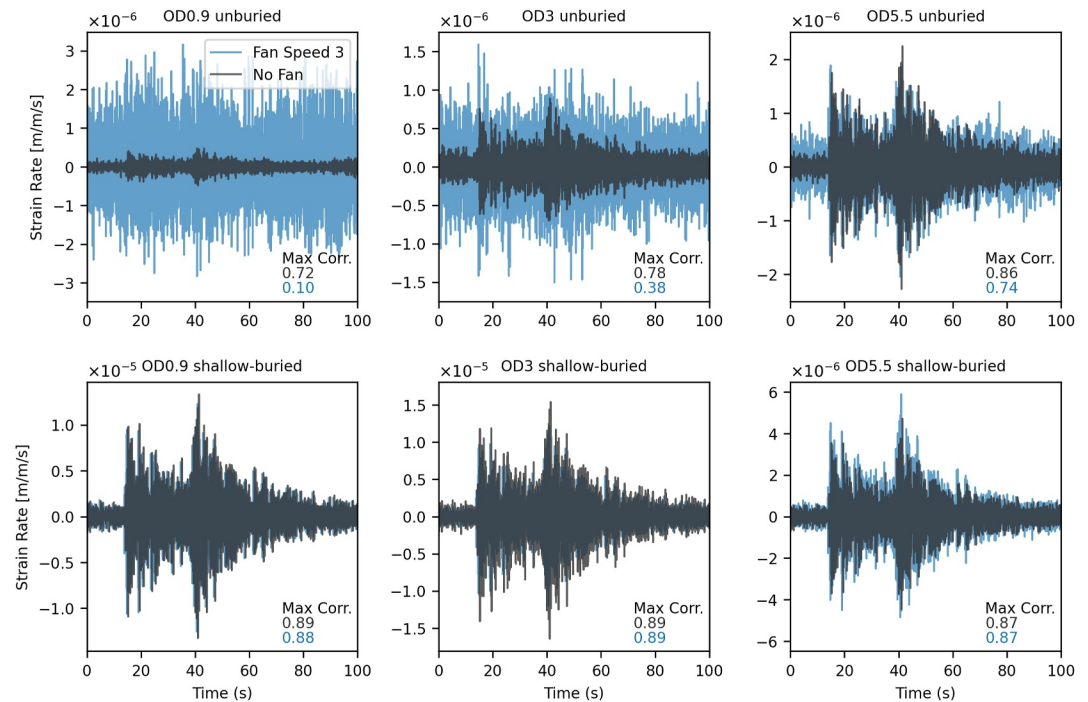


Figure 5. Comparison of simulated earthquake signals recorded on different fiber segments with and without wind noise. The signals were generated with the shaker to simulate a seismic event. Black lines show recordings in the absence of any fan operation (i.e., no wind and no fan motor noise), while blue lines show recordings with the fan operating at fan speed 3 corresponding to a wind speed of 3 m/s. All traces were bandpass filtered between 3 and 15 Hz. The simulated earthquake is clearly visible in all traces without wind, although with lower signal strength on OD0.9 unburied and OD3 unburied. With wind present, the seismic signal becomes completely obscured in the OD0.9 and OD3 unburied traces, while it remains visible but with reduced SNR on the OD5.5 unburied trace. All shallow-buried fiber segments show minimal effect from wind noise. Maximum correlation values (bottom right of each panel, gray: No Fan, blue: Fan Speed 3) quantify the signal fidelity by comparing recorded signals to the driving waveform, with higher values indicating better signal-to-noise ratio. The correlation values confirm that shallow-buried segments maintain high signal fidelity regardless of fan operation, while unburied segments show significant degradation with wind, except for OD5.5, which maintains moderate correlation.

The SNR analysis supports these findings (Figure 2d). For OD5.5 and OD3, SNR values were similar between buried and unburied deployments, indicating that signal detectability remains robust even without burial. In contrast, OD0.9 showed substantial SNR reduction when unburied, consistent with its lower amplitude and higher variability. These results align with the findings of Zandanel et al. (2026), who observed reduced amplitudes but comparable SNR for unburied OD3 compared to buried OD3 in earthquake recordings. The noise in our experiment is predominantly environmental and not instrumental: Since the noise itself is influenced by the same coupling mechanisms as the signal, the SNR trends partly reflect signal-to-signal comparisons rather than signal-to-instrumental-noise ratios. In a quieter environment like the lunar surface, the relative SNR differences between cable types may be more pronounced.

Continuous ground contact is critical in unburied deployments. Surface undulations causing free-hanging segments degrade signal quality, leading to unreliable amplitudes and phases (Figure 3). Even short segments (~20 cm) negatively affected all cables, with impact increasing with undulation length. These segments likely resonate when excited, with resonance frequencies depending on cable properties and segment length, supported by observed phase shifts. Resonance-like behavior was observed for the unburied OD0.9 even in the absence of undulations, suggesting that this thinner cable is particularly prone to resonating when not well coupled. Resonance effects in DAS cables have also been reported in previous studies, such as Hudson et al. (2021), who observed wind-induced cable resonance in unburied deployments.

Deployment technique significantly influences signal quality. Pressing cables down to ensure ground contact reduces free-hanging segments and improves coupling. This aligns with findings by Mjehovich et al. (2023), who observed improved signal quality when cables were pressed into snow compared to deployment without further

coupling efforts, and by Harmon et al. (2022), who reported poor coupling on grassy surfaces where cables were suspended above the ground by the grass.

Wind noise significantly increases the noise floor for unburied cables (Figure 4). The extent to which cables were affected correlates with their diameter and mass: the lighter OD0.9 was most susceptible to wind induced noise, followed by the OD3, then OD5.5. However, all cable types exhibited wind noise amplitudes of over 10^{-7} m/m/s in the 0–20 Hz frequency band, a significant contribution to the noise floor and strong enough to mask earthquake signals that would in quiet conditions be well above the threshold for detection. This suggests that wind is likely a major contributor to the poor signal quality often observed in unburied deployments on Earth, consistent with other studies that have identified wind as a dominant noise source in surface DAS deployments (Carr et al., 2025; Hudson et al., 2021; Viens et al., 2025).

The experimental results for OD5.5, representative of typical terrestrial DAS cables, provide insight into the dominant factors affecting signal quality in unburied terrestrial deployments. This cable maintained stable amplitude and phase even when unburied throughout the tested frequency range, with only minor degradation compared to its shallow-buried configuration. Wind did lead to a significant increase in the noise floor, but the cable remained capable of detecting strong seismic events. In contrast, surface undulations had a strong detrimental impact: already relatively short (15–25 cm) free-hanging segments led to significant amplitude reduction and erratic phase behavior, particularly at higher frequencies. These results suggest that unburied DAS can achieve signal quality comparable to buried configurations in the absence of strong wind and if continuous ground contact is maintained. The feasibility of this depends on the deployment environment, but strategies such as pressing the cable down or flattening the surface could be employed to increase ground-cable contact.

4.2. Coupling Mechanisms

Our results show that signal quality in unburied deployments varies strongly with cable type and deployment details. Thinner cables and configurations with reduced ground contact consistently yielded lower amplitudes and less reliable phase information. While these empirical patterns are clear, the physical mechanisms that govern strain transfer from the ground to the optical fiber remain poorly understood. Understanding coupling mechanisms is crucial for generalizing results, predicting performance under lunar conditions (e.g., reduced gravity), and designing optimized cables. Strain from ground motion is first transferred from the ground to the cable, and then from the cable to the fiber. Reinsch et al. (2017) developed a theoretical model for cable-to-fiber strain transfer in multilayer fiber-optic cables, showing that the efficiency of strain transmission from the cable surface to the optical fiber depends on cable geometry and material properties, and varies with wavelength. Their analysis highlights that strain transfer improves with increasing wavelength and is nearly ideal for typical seismic wavelengths. The ground-to-cable transfer depends on the contact points between ground and cable, the cable geometry in between contacts, and the relative impedance of the ground and cable, all which depend on the mechanical properties of the ground surface and the cable.

In buried configurations with tight ground contact and no slip, coupling is assumed to be ideal—the cable moves with the ground. In that case, the strain transfer from the ground to the fiber is primarily determined by the cable-to-fiber transfer. In unburied setups, however, the cable only follows ground motion at contact points, but not necessarily on the segments in between. Assuming that burial provides good ground-cable coupling, the observed differences in average amplitude ratios between buried cable types (Figure 2c) primarily reflect variations in cable-to-fiber transfer efficiency. The OD0.9 exhibits approximately double the average amplitude ratio compared to the OD3 and OD5.5 when buried, indicating more efficient strain transfer through its cable layers. For the unburied cables, the observed amplitude variations result from the combined effects of both ground-to-cable and cable-to-fiber transfer steps. The OD0.9, despite showing the highest amplitudes when buried, exhibits very weak amplitudes when unburied, suggesting poor ground-cable coupling. The OD3 shows a smaller reduction from buried to unburied conditions, indicating somewhat better ground-cable coupling. Notably, the OD5.5 maintains similar average amplitude ratios regardless of burial condition, suggesting effective ground-cable coupling even without burial. Our findings are consistent with the numerical simulations by Celli et al. (2023), who modeled DAS cables using a particle-spring lattice and demonstrated that weak ground-cable coupling can cause significant amplitude variations and phase delays. Their study also showed that cable material properties have a non-negligible effect on the recorded response.

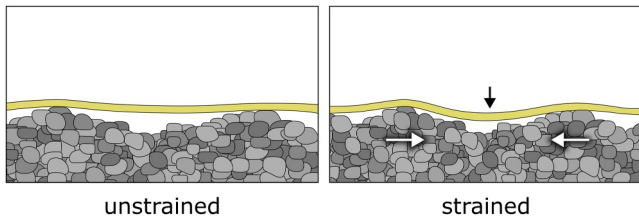


Figure 6. Conceptual illustration of the bending stress relief mechanism proposed to explain reduced strain transfer in unburied DAS cables. When ground deformation causes contact points to move closer or further apart, suspended cable segments bend rather than compress or stretch, relieving stress and reducing axial strain transmission to the fiber core.

We hypothesize that the main ground-cable coupling mechanism in unburied deployment is bending stress relief described in the following. In our unburied deployment, cables are draped as flat as possible, yet typically have a few mm to cm between ground contact points. Between contacts, the cables are suspended in the air due to their stiffness and slightly arching toward the ground due to gravity. When the ground deforms and neighboring contact points move toward each other, the cable segment suspended between them can bend perpendicularly to the strain direction (as illustrated in Figure 6). This bending relieves part of the stress rather than transferring it as axial strain to the fiber core. While a full quantification of this effect is beyond the scope of this study, beam theory suggests that the resistance to bending increases with both the Young's modulus and the fourth power of the cable diameter. This implies that thinner and more flexible cables bend more easily under hori-

zontal shortening, thereby relieving more stress through bending and reducing the strain transferred to the fiber. This aligns with our experimental observations, where thinner unburied cables consistently recorded lower amplitudes. In addition to cable stiffness and diameter, the length of the suspended segment also plays a critical role: longer segments bend more easily under the same horizontal deformation, further reducing strain transfer. This aligns with our observation of decreased amplitudes for all cable types with increasing undulation length. When free-hanging segments become too long, resonance effects may dominate, further degrading strain transfer and introducing phase shifts.

Slip between the ground and the cable is another potential mechanism affecting coupling. Slip occurs when the force required to deform the cable exceeds the frictional force at the contact points. If slip were to occur, it would be amplitude-dependent: small deformations could be sustained by friction, while larger deformations might exceed the frictional threshold, leading to slip. This would imply a non-linear coupling behavior at the onset of slip. To assess whether slip is likely under realistic seismic conditions, we compared the frictional force at the ground-cable contact to the deformation force required to deform the cable. The frictional force per unit length is estimated as $F_f/L = \mu \cdot m_c \cdot g \approx 0.07 \text{ N/m}$, where $\mu \approx 0.5$ is the friction coefficient which can vary depending on the surface properties, $m_c = 0.015 \text{ kg/m}$ is an average cable mass per unit length, and g is gravitational acceleration. The deformation force per unit length, calculated using Hooke's law and assuming strain amplitudes typical of seismic waves, is given by $F_d/L = \frac{2AE\epsilon_{\max}}{\lambda}$, where A is the cross-sectional area of the cable, E is the Young's modulus, λ is the wavelength of the seismic signal, and $\epsilon_{\max}$ is the peak strain amplitude. For strain amplitudes typical of natural earthquakes, this expression yields $F_d/L \approx 0.0012 - 0.012 \text{ N/m}$, while for stronger, active sources, this could increase to the range of $0.05 - 0.5 \text{ N/m}$. To summarize the slip condition, we define a dimensionless number comparing the available frictional force to the required deformation force:

$$\Gamma = \frac{F_f/L}{F_d/L} = \frac{\mu \cdot m_c \cdot g \cdot \lambda}{2AE\epsilon_{\max}}, \quad (1)$$

where $\Gamma > 1$ indicates that friction is sufficient to prevent slip. For the parameter ranges considered, $\Gamma > 1$, suggesting that slip is unlikely under most realistic seismic scenarios. This conclusion is consistent with our experimental observations: we did not detect any non-linear behavior indicative of slip, even at amplitudes larger than those expected from natural seismic events. However, slip may still be relevant under conditions with low friction, such as due to substrate or cable properties, or under reduced lunar gravity, which lowers the normal force and thereby the available friction. Harmon et al. (2024) hypothesize that slip may have occurred in their unburied DAS experiment due to the low-friction properties of the cinder and ash substrates, and suggest that such slip can act like attenuation and cause frequency-dependent phase shifts.

4.3. Implications for Lunar DAS

Our results indicate that unburied DAS may be viable for lunar applications, especially with robust cables like the OD5.5, which exhibited reliable strain transfer in the tested frequency range. However, the detrimental effects of free-hanging segments underscore the importance of careful deployment. We recommend unspooling the cable onto the surface and pressing it down, for example, by rolling a wheel over it. This simple method could significantly enhance coupling, as even minor deployment variations had substantial effects in our experiments. A

similar concept has been proposed by Burns et al. (2023), who suggest rolling out a flat tape tether for lunar deployment. The feasibility of such deployment strategies depends on cable length and mission constraints. Zhai et al. (2024) investigated optimal DAS array lengths for various scientific objectives: cables exceeding 10 km would enable deep interior studies, while shorter deployments of hundreds of meters would suffice for characterizing thermal moonquakes and local shallow structure. Short cables could potentially be deployed using launch mechanisms that project the cable from the landing site. Longer deployments would require robotic rovers, which could facilitate the pressing-down procedure during cable deployment. Our results also show that wind noise is a major factor degrading signal quality in unburied DAS on Earth. This issue would not be present on the Moon due to the absence of an atmosphere, representing a clear advantage for lunar deployments. While other potential noise sources exist on the Moon, such as regolith transport and micrometeorite impacts, these are intermittent phenomena and unlikely to pose the same continuous challenges as terrestrial wind. Characterizing these intermittent signals with DAS could even provide valuable insights into lunar surface processes, though further studies are needed to predict their expected signatures in DAS data.

An important consideration for lunar deployment is the effect of reduced gravity on ground-cable coupling. We observed that heavier and stiffer cables, such as the OD5.5, exhibited better coupling quality. If cable weight is indeed a dominant factor in achieving good coupling, then replicating this behavior on the Moon, where gravity is approximately one-sixth that of Earth, might require cables that are six times heavier to exert the same normal force on the ground. However, it remains unclear whether weight alone governs coupling quality. Other factors such as cable diameter and stiffness may play equally or more significant roles, particularly if bending stress relief is a relevant factor. Since these properties are independent of gravity, they would remain unchanged on the Moon. This uncertainty highlights the need for a deeper understanding of the physical mechanisms behind strain transfer.

5. Conclusion

We investigated the coupling behavior of fiber-optic cables deployed in a lunar regolith simulant, with a focus on assessing the feasibility of DAS for lunar seismic applications. We systematically evaluated the impact of cable type, cable burial, surface undulations, and wind noise on signal quality. Buried cables consistently recorded reliable phases and higher amplitudes. Unburied cables, while exhibiting lower absolute amplitudes, maintained stable amplitude ratios and correct phases across the tested frequency range, except for the thinnest cable (OD0.9), which showed poor coupling and was highly sensitive to wind noise and deployment details.

Our results suggest that coupling quality in unburied deployments is governed by mechanical factors such as cable stiffness, diameter, and ground contact continuity. We proposed bending stress relief as a key mechanism explaining reduced strain transfer in unburied cables, supported by both theoretical considerations and experimental observations. No evidence of nonlinear coupling behavior was found, indicating that slip between cable and ground is unlikely under typical seismic conditions.

These findings have direct implications for lunar DAS deployment. One of the major limitations of unburied DAS on Earth is atmospheric noise, particularly wind, which significantly degrades signal quality. The absence of a dense atmosphere on the Moon removes this constraint, making unburied DAS a promising option for lunar applications. Our results suggest that thicker cables like the OD5.5 can maintain reliable coupling even without burial, provided that continuous ground contact is ensured and free-hanging segments are avoided. Simple strategies such as pressing the cable onto the surface during deployment could substantially improve coupling.

An important difference between our experimental setup and the lunar environment is the reduced gravity, which directly affects the weight of the cable and thus its contact force with the ground. Future research should aim to better understand the physical coupling mechanisms quantitatively to assess how reduced gravity influences strain transfer. Modeling studies or reduced-gravity experiments could help clarify these relationships and guide the design of optimized DAS cables for lunar deployment.

Conflict of Interest

The authors declare no conflicts of interest relevant to this study.

Availability Statement

The used data sets and data processing codes are openly available at the ETH Zurich Research Collection via <https://doi.org/10.3929/ethz-c-000785214> (Probst et al., 2025).

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